

Calculus II quiz 1 第十章选择填空题

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) The series $\sum_{n=1}^{\infty} (1 - \frac{a}{n})^{n^2}$ converges ()

- A. for any real number a . B. A. for any $a \in (-1, 1)$.
C. for any $a > 0$. D. for any $a < 0$.

(2) If $0 \leq a_n \leq \frac{1}{n} (n = 1, 2, \dots)$, which of the following series must be convergent ? ()

- A. $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{(1+\ln n)^2}$. B. $\sum_{n=1}^{\infty} (-1)^n a_n$. C. $\sum_{n=1}^{\infty} a_n$. D. $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{1+\ln n}$.

(3) If $\sum_{n=1}^{\infty} a_n x^n$ converges at $x = -2$, then ()

- A. $\sum_{n=1}^{\infty} a_n 2^n$ converges. B. $\sum_{n=1}^{\infty} a_n 2^n$ diverges.
C. $\sum_{n=1}^{\infty} n a_n$ converges. D. $\sum_{n=1}^{\infty} n a_n$ diverges.

(4) If $\sum_{n=1}^{\infty} a_n$ converges, then ()

- A. $\sum_{n=1}^{\infty} |a_n|$ converges. B. $\sum_{n=1}^{\infty} (-1)^n a_n$ converges.
C. $\sum_{n=1}^{\infty} a_n a_{n+1}$ converges. D. $\sum_{n=1}^{\infty} \frac{a_n + a_{n+1}}{2}$ converges.

(5) When $\lim_{n \rightarrow \infty} a_n = 0$, then ()

- A. If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n b_n$ converges.
B. If $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n b_n$ diverges.
C. If $\sum_{n=1}^{\infty} |b_n|$ converges, then $\sum_{n=1}^{\infty} a_n^2 b_n^2$ converges.
D. If $\sum_{n=1}^{\infty} |b_n|$ diverges, then $\sum_{n=1}^{\infty} a_n^2 b_n^2$ diverges.

(6) Let $a_n > 0$ for all n . Which of the following statements must be true ? ()

- (A) If $\lim_{n \rightarrow \infty} n a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
(B) If $\lim_{n \rightarrow \infty} n a_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
(C) If $\lim_{n \rightarrow \infty} n a_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
(D) None of the above statements is correct.

(7) Let a be a constant, the series $\sum_{n=2}^{\infty} (\frac{\sin(n+a)}{n^{1.01}} - \frac{1}{n \ln n})$ ()

- (A) converges absolutely. (B) converges conditionally. (C) diverges.
(D) converges or not depending on the value of a .

(8) The interval of the convergence for the power series $\sum_{n=1}^{\infty} \frac{x^n}{n 3^n}$ is ()

- (A) $[-\frac{1}{3}, \frac{1}{3}]$. (B) $[-\frac{1}{3}, \frac{1}{3})$. (C) $[-3, 3]$. (D) $[-3, 3)$.

- (9) Let a be a constant. Then the series $\sum_{n=1}^{\infty} \left(\frac{\sin(an)}{n^2} + \frac{(-1)^n}{n+1} \right)$ ()
 (A) converges absolutely. (B) converges conditionally. (C) diverges. (D) converges or not depending on the value of a .

- (10) Which one of the following series diverges? ()
 (A) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. (B) $\sum_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{n}}}$. (C) $\sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}}$. (D) $\sum_{n=1}^{\infty} \frac{(-1)^n (3+(-1)^n \cdot 2)^n}{6^n}$.
 (11) Suppose $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), then which of the following series converges ()
 (A) $\sum_{n=1}^{\infty} a_n$. (B) $\sum_{n=1}^{\infty} (-1)^n a_n$. (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$. (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

2. Fill in the blanks.

- (1) If $\sum_{n=1}^{\infty} \left(\tan \frac{1}{n} - k \ln \left(1 - \frac{1}{n} \right) \right)$ converges, then $k =$ ().
 (2) $\lim_{x \rightarrow 0} \frac{3 \sin(2x) - 2 \sin(3x)}{6x^3 + x^4} =$ ().
 (3) $\lim_{x \rightarrow 0} \frac{\sin x - x}{(\cos x - 1)(e^{2x} - \cos x)} =$ ().
 (4) The sum of the series $\frac{1}{2} + \frac{1}{4 \cdot 2!} + \frac{1}{8 \cdot 3!} + \frac{1}{16 \cdot 4!} + \dots + \frac{1}{2^n \cdot n!} + \dots$ is ().
 (5) The series $S = -\ln 2 + \frac{\ln^2 2}{2!} + \dots + \frac{(-1)^n \ln^n 2}{n!} + \dots$ converges and has the sum ().
 (6) If the Maclaurin series of a function $f(x)$ is $\sum_{n=1}^{\infty} \frac{x^{2n+2}}{n(n+4)2^n}$, then $f^{(6)}(0) =$ ().

3. Determine whether the following statements are true or false? No justification is necessary.

- (1) If $a_n < 0$ for $n > 1000$, and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ may diverge.
 (2) Suppose that the power series $\sum_{n=1}^{\infty} a_n(x-1)^n$ converges at $x = 0$, and diverges at $x = 2$, then the interval of convergence of this series is $[0, 2)$.